Isaiah Coleman

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Engineering Manager | DevOps & Platform Engineering | Expert in Building for Scale & High-Growth Startups

Skills

Leadership & Management: Team Leadership , Project Management , Agile Development , Hiring & Mentoring , Cross-functional Collaboration
Languages: Node.js/TypeScript, Java - **10+ years** , Rails - **6+ years** ,Elixir ,Python - **3+ years**, Go - **1+ years**
Frontend Frameworks: React - **7+ years**, Angular - **2+ years**, Hotwire (rails html-over-the-wire) - **2+ years**, Redux, jQuery, Javascript
Cloud & Infrastructure: AWS - **6+ years**, GCP - **2 years**, Azure **2 years**, Kubernetes - **5+ years**, Openshift Kubernetes(OKD) - **2+ years**, Docker, Terraform, Ansible, Puppet, Jenkins, CI/CD Pipelines, AWS Lambda, AWS EC2/ECS, Cloudformation, Github Actions, Gitlab CI/CD, Serverless, Microservices Architecture
Testing: Jest, Capybara, RSpec, ExUnit, JUnit, Selenium
Containerization & Orchestration: Docker, Podman, Docker Compose, Kubernetes
Development Tools & Practices: Git, Bash, Unit Testing, JIRA, Agile, Kanban, Scrum
AI & LLM Chabot Technologies: Ollama, Claude, Codex, Continue, LiteLLM/Open AI API, MCP
Databases & Data Storage: Postgres, Redis, Pubsub, Redshift, Kinesis
Identity & Access Management (IAM): Okta, Cognito, SSO, OAuth 2.0, SAML 2.0
Security & Compliance: Federal/International Law - FERPA / COPPA / GDPR, WCAG, A11Y, CVE Vulnerability patching, Trivy Static Container Vulnerability Scan

Experience

Duke University Libraries, Senior *Web Application Developer (Rails Lead & platform engineer)*

2024-present

- Lead developer of ruby on rails fork of Hyrax open source repository system. Participating in the samvera community, contributing upstream patches, and developing bespoke features for the universities research data repository system
- Orchestrate development lifecycle and project planning of various projects working with a distributed team of domain experts, across different 3+ countries and 4+ organizations
- Contribute to upstream open source community projects submitting multiple features and patches a year
- Migrate legacy Linux VM based applications to containerized openshift kubernetes services reducing deployment time and by 70% and improving consistency across environments
- Maintain hybrid (on-prem and cloud) infrastructure as code with Ansible, Helm supporting 12+ services
- Implement CI/CD pipelines adding automatic deployment, test automation, code-linting, and static container vulnerability analysis
- Lead conversations regarding build versus buy versus FOSS. For example, researching and deploying FOSS bot-blocking utility to handle sporadic +7,000% request/second increases in traffic and saving greater than $10k a year
- Build LLM-based applications using MCP, reducing manual research/metadata tasks by an estimated 25% for staff

NCSU University Libraries, *Web Application Developer(PHP Lead)*

2023 - 2024

- Led containerization of legacy applications, reducing resource usage by ~40% and deployment time by >60%.
- Refactored monolithic legacy applications into microservices using modern frameworks ensuring their longevity and maintainability and reducing technical debt backlog by 50%
- Maintained Hybrid Cloud infrastructure using Ansible and Puppet supporting 10+ services
- Developed open source library used to maintain EZProxy web stanzas and maintained campus-wide online periodical proxying service supporting hundreds of thousands of daily requests

Crosscomm, *Lead Backend Developer(Next.js & AWS Cloud Infrastructure Lead)*

2022-2023

- Architected NextJS services serving dually as team lead for all backend development, heading all SRE/Operations and infrastructure responsibilities for AWS infrastructure
- Implemented Node.js/Typescript microservices with AWS Lambda and API Gateway
- Led development of AWS based high-performance-computing cluster 'manager' for Duke research labs. Providing an interface empowering their team to effortlessly create, modify, and deploy slurm workloads to high-performance computing clusters designed to handle GPU-intensive jobs for scientific research endeavors.
- Maintained Cloud Native infrastructure in AWS and Azure with Terraform
- Instituted minimum coding standards guidance and implemented code-linting, and unit-testing through github CI/CD decreasing deprecated feature and defect rates by 30%
- Led client communication and orchestrated development lifecycle, from requirements gathering to system architecture design

LearnPlatform (Instructure)**,** *Senior Integration Engineer*

2019 - 2022

- Led development of all third party (6+) integrations and designed, implemented, and deployed the strategy (and security policies) through which all third party data was consumed by way of an Elixir backend
- Architected interoperable SIS data ingest into Elixir backend processing data for 2+ million students and teachers into Elixir Backend
- Implemented Node.js/Typescript microservices in AWS Lambda using API Gateway, Pub/Sub and other invoke triggers supporting 700k+ daily messages and 90m+ daily check-in requests from district devices.
- Ensured stability of Elixir backend handling 30+ billion yearly published events
- Served as scrum lead, technical sales consultant, and hiring manager. Screened 200+ candidates, performed 30+ technical interviews and helped hire the next 10 engineers.
- Developed in-house engineering practices and ceremonies for unit testing, linting, containerization, standups, code review practices, release planning, and test plans. Progressed team from large waterfall monthly deliveries to a weekly release schedule.
- Managed AWS ECS infrastructure ensuring platform stability to handle increasing load on the system.
- Enshrined Cloud Native infrastructure in AWS and GCP in code using Terraform
- Performed full-stack development with React frontend and Ruby On Rails backend

SAS, *Fullstack Web Developer*

2013 - 2019

- Conducted full-stack software development on Java Web backend and redesigned existing JSP frontends to utilize modern javascript frameworks refactoring from jQuery to React
- Developed java libraries to consume and export metadata in conformance with open EdTech standards: One Roster, Common Cartridge, Open Badging, and Caliper used for interoperability across 40+ integrations with school districts and third-party organizations
- Automated in-house conformance testing, reducing annual testing time by ~70%
- Implemented java web SSO integrations, utilizing open source SAML 2.0, OAuth 2.0, and LTI libraries and folding them into in house authentication and access library supporting 4+ million users
- Ensured conformance to federal Child Privacy laws (FERPA, COPPA) developing automated anonymization procedures and defining risk profiles for all collected data

Education

North Carolina State University, Raleigh — *B.S. Computer Science* 2009 - 2013